Threshold Post-Quantum Signing, and the Holder Side with Longfellow

A short conversation deck

Changqi Sun
PhD student, Virginia Tech (advisor: Thang Hoang)

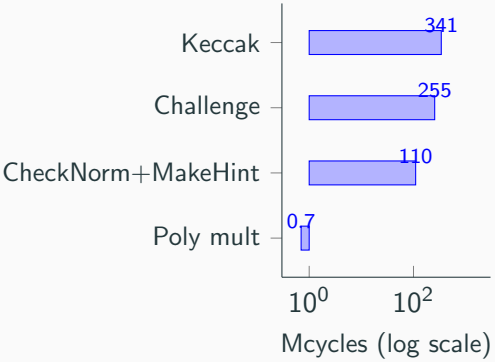
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About me

- Fourth-year PhD student at Virginia Tech, applied cryptography; graduating 2028.
- Main line: **threshold signing for lattice signatures**, so that no single server holds the key and the output is a standard signature.
  - **SHIELD** (IEEE S&P 2026, co-first author): threshold ML-DSA.
  - **FLOCK** (first author, under submission): Falcon-compatible threshold signatures.
- Also ORAM: **HERO** (first author, under review), distributed ORAM with an untrusted helper.

Common thread: compute and prove things about unmodified standard signatures.

SHIELD: unmodified ML-DSA signing inside MPC (online, ML-DSA-44)



- n servers, malicious security; output bit-identical to FIPS 204 (NIST KATs).
- SPDZ + WRK17 garbled circuits, daBits to switch.
- NTTs are linear on shares: $< 0.2\%$ of online compute; Boolean work dominates.
- Verilog $\rightarrow$ Yosys $\rightarrow$ Bristol; SampleInBall 890K $\rightarrow$ 355K gates.

SHIELD cost table (signing in MPC). ML-DSA-65/87: Keccak 438/597, poly 1.21/1.37 Mcycles.

FLOCK and the threshold-Falcon fixes

FLOCK (first author, under submission)

- Falcon-compatible threshold signatures: the Falcon counterpart of SHIELD.
- Together: threshold signing for both lattice signatures NIST selected.

Before FLOCK: reproducing a CCS 2026 artifact

- Found two silent bugs: a PRG byte/element mix-up left 7/8 of the sampler buffer uninitialized; a `size_t` swallowed -1.
- Fixes merged upstream; signing $\approx 9\times$ faster.

What this says about me: I read other people's cryptographic code closely, and I check it against the spec.

My reading of Longfellow

The statement

- An mdoc (ISO 18013-5) credential signed with ECDSA P-256 + SHA-256: prove possession and reveal one attribute, with no change to the issuer.

The recipe

- Run sumcheck, commit to the padded transcript, then prove "the transcript makes the verifier accept" with Ligero.
- Ligero only sees $d < 20$ quadratic constraints, not the whole circuit.

The fields

- P-256 base field: NTT in a quadratic extension, $\approx 1.5\times$ cost.
- SHA-256 in $\mathrm{GF}(2^{128})$, additive FFT.
- A random MAC links the witnesses of the two proofs.

Today: ECDSA prover 16.7 ms (M4); CFRG draft; part of the PQCA.

Frigo & shelat, CiC 2025 (ePrint 2024/2010); draft-irtf-cfrg-longfellow-00; IETF 126 slides.

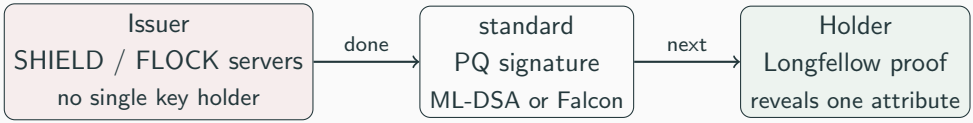
Your ML-DSA circuit, through the lens of SHIELD

FIPS 204 Alg. 8 step	kind	Keccak-f (ML-DSA-44)
$\hat{A} \leftarrow \mathrm{ExpandA}(\rho)$ (SHAKE128)	hash	≈ 96
$tr \leftarrow H(pk)$	hash	10
SampleInBall, $\tilde{c}' = H(\mu \,\|\, w_1')$	hash	≈ 10
$\mathrm{NTT}^{-1}(\hat{A} \circ \hat{z} - \hat{c} \circ \widehat{t_1 2^d})$	mod q	–
UseHint, $\|z\|_\infty < \gamma_1 - \beta$	range	–

- In `tests/pq/ml_dsa`, $(\hat{A}, t_1, tr)$ are public inputs, so ExpandA and $H(pk)$ stay outside: ≈ 10 permutations. **858 ms** on an M4 (IETF 126).
- Same picture as signing in SHIELD: the lattice algebra is cheap, the hashing is not.
- If the public key must be hidden: ≈ 116 permutations, ≈ 4.5M AND gates.

FIPS 204; 16 ExpandA polynomials $\times \leq 6$ SHAKE128 blocks; $pk = 1312$ B over a 136 B rate; Keccak-f[1600] = 38,400 AND gates.

What comes next: from the issuer side to the holder side



B. ML-DSA end to end

- SHIELD signs an ML-DSA credential; your existing circuit proves it.
- No change to the verifier or the circuit: the signature is standard.
- Then: cost of the whole pipeline, issuer and holder.

F. A Falcon circuit for Longfellow

- FLOCK signs Falcon credentials; the holder proves possession.
- Falcon verification is short, and hiding the key is almost free (next slide).

Why Falcon is interesting for the holder side

Falcon verify (round-3 spec, Alg. 16)

1. $c \leftarrow \mathrm{HashToPoint}(r \,\|\, m)$: SHAKE256, 16-bit samples, keep $t < 5q$.
2. $s_2 \leftarrow \mathrm{Decompress}(s)$.
3. $s_1 \leftarrow c - s_2 h \bmod q \quad (q = 12289,\ n = 512)$.
4. Accept iff $\|(s_1, s_2)\|^2 \le \lfloor \beta^2 \rfloor$.

The public key is one polynomial h: nothing to expand.

Keccak-f per verification (my estimate)

	key public	key hidden
ML-DSA-44	≈ 10	≈ 116
Falcon-512	≈ 9	≈ 9

Open questions: Decompress in the circuit or as a witness; which field for $q = 12289$ ($q - 1 = 3 \cdot 2^{12}$, so a 512-point negacyclic NTT exists).

Falcon spec v1.2 (round 3); FIPS 206 is still a draft. 512 samples $\times$ 2 B / 0.94 acceptance $\approx$ 1.1 KB over a 136 B rate.

Questions for you

1. What is next for post-quantum Longfellow: issuer-side ML-DSA in mdoc, or device binding first?
2. What would make the ML-DSA circuit production-grade: speed, formal verification, or the spec?
3. Is Falcon on your radar, or is ML-DSA the only PQ target for now?
4. Where would an intern's code land: the C++ library, the Rust rewrite, or the formal side?

One more thing

I'm already in the PhD SWE intern pool.

If you think there is a fit, would you be open to requesting me from the pool for your team?

I'm happy to send a one-page project sketch this week.

Thank you!

changqi@vt.edu changqi.dpdns.org